1 Product Introduction

The electro-optical identification and tracking device integrates an HD visible-light camera, cooled/uncooled thermal imager, laser illuminator, multispectral detection modules, automatic tracking, intelligent analytics, and precision electronic control. With a compact concealed design, it detects, locates, tracks, and identifies small UAV targets, and can cross-cue with communication detection and radar detection subsystems to verify aerial small UAVs or other low-slow-small targets and collect video evidence.

The high-quality housing is made of ultra-high-strength aluminum alloy with a spherical low-drag design for strong wind resistance and low vibration. Dual internal and external protective treatment and vacuum nitrogen sealing provide an IP66 protection rating, ensuring long-term stable operation in harsh outdoor environments.

Figure 1 Electro-Optical Identification and Tracking Device

2 Key Features

- High Payload Capacity

Can carry a long-focus visible-light camera, large-aperture thermal imager, and tracking laser illuminator. Optional laser ranging, positioning/navigation, digital compass, and other sensor modules enable ultra-long-range target observation.

- Wide Detection Spectrum

Can integrate HD visible-light imaging, cooled/uncooled thermal imaging, and laser illumination. Multi-band detection advantages complement each other, combining active and passive detection with multisource data fusion to meet day/night and all-weather monitoring requirements.

- Fast Slewing

Slewing speed can reach 80°/s and acceleration can reach 100°/s^2. Rapid start/stop and smooth motion help capture and track fast-moving targets.

- Wide Coverage

Azimuth rotation range is 0°-360° and elevation rotation range is -90° to +90°, enabling blind-spot-free detection and full-dimensional coverage.

- High Control Accuracy

Uses a precision angle encoder together with a high-accuracy closed-loop servo control system, providing positioning accuracy up to 0.02°. A high-performance image processing unit and precision focus mechanism enable accurate autofocus.

- Excellent Tracking Performance

The automatic tracking module uses multiple advanced target acquisition and tracking algorithms, together with high-precision servo control, ensuring stable tracking during rapid flight and direction changes.

- High Intelligence

Supports abnormal behavior detection such as target intrusion and motion detection, as well as target classification recognition, hotspot alarms, radar linkage, 3D zoom positioning, and other functions, greatly improving unmanned, automated, and information-driven operation.

- Multiple Active Protection Measures

Intense heat source protection, internal temperature monitoring, and a directional industrial defroster improve system reliability.

- Strong Environmental Adaptability

High-strength aluminum alloy materials, dual internal and external protective treatment, and IP66 protection.

Basic Specifications	
Operating Range for a 0.3 m × 0.3 m UAV Target	Visible-light tracking range ⩾4,000 m; identification range ⩾3,000 m. Thermal tracking range ⩾2,000 m; identification range ⩾800 m.
Spectral Detector	5th-generation uncooled VOx vanadium oxide detector, 640×512 resolution, 22-230 mm lens, 10× continuous optical zoom
Visible-Light Detector	2 MP HD camera core, 12.5-775 mm HD motorized zoom lens, day/night color-to-monochrome CMOS
Intelligent Analytics	Boundary intrusion, automatic tracking, UAV identification
Heavy-Duty Pan-Tilt Unit	0.01° high-precision positioning; azimuth rotation range 0-360°; elevation rotation range ±90°; slewing speed 80°/s; minimum speed 0.01°/s

IP Network Interface	Supports ONVIF/RTSP/28181 protocols
Power Supply	AC 220 V
Total Power Consumption	⩽500 W
Overall Dimensions	Φ565 mm × H880 mm
Total Weight	⩽105 kg